AgentCgroup

AI Agent Resource Characterization and OS-Level Control

Yusheng Zheng (yzhen165@ucsc.edu)
PhD Student

Roadmap

- Motivation & Problem
- Resource Characterization (144 tasks, 2 models)
- Agent vs. Cloud Workloads & Three Mismatches
- AgentCgroup: eBPF + Bash Wrapper + Negotiation
- Evaluation & Takeaways

Motivation

AI coding agents **Claude Code, Codex, Devin** are being deployed at scale, autonomously running compilers, test suites, and package managers in sandboxed containers.

New Workloads: AI Agents

- LLM reasons → calls tools → observes → iterates
- Agent **autonomously decides** what to run and when
- Peform actions 24/7, tools may be heavy and spike

…How much does it actually cost?

- People Host **50–100+ agents per machine**
- Does traditional **containers** and **VMs** enough?
- Most study around Safety, what about **resource usage** and **performance**?

Characterization: Setup

144 SE tasks from SWE-rebench benchmark, with Claude Code

Config	Model	Tasks
Local GPU	GLM-4.7-Flash	111
Cloud API	Claude Haiku 4.5	33

Hardware: Intel Core Ultra 9 285K, 24 cores, 128 GB DDR5

Environment: Podman containers, Linux 6.15, 1s sampling

What We Measure

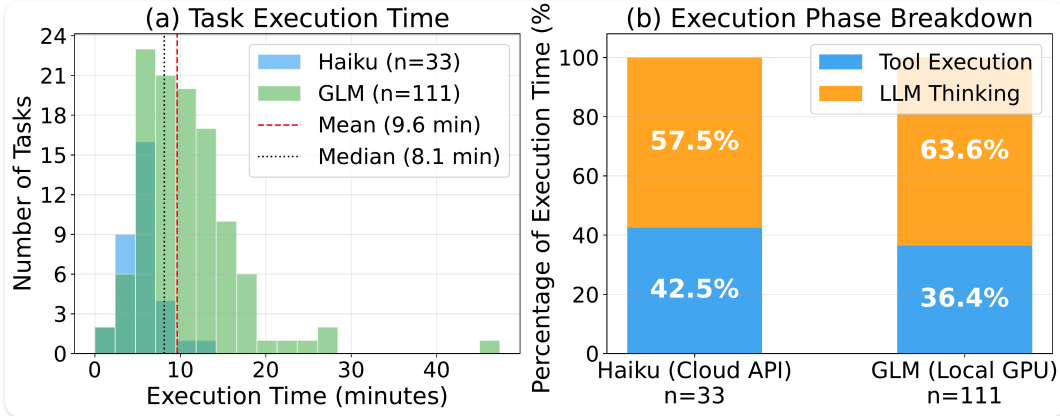
- **Where does agent do?** (execution model)
- **What does resource usage look like?** (resource dynamics)
- **Can demands be predicted?** (cross-task variance)

Container Images

- **Average 3.5 GB, 7× larger than microservice and 70× larger than FAAS**

Execution Model

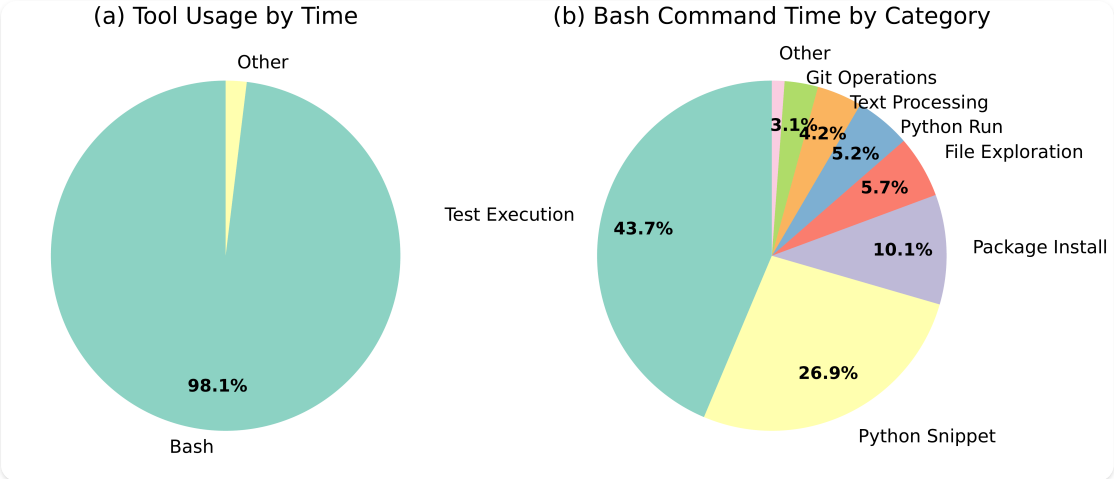
Q: Where does the time go?



- Task duration: mean 9.6 min
- LLM reasoning: 26–44% of end-to-end task latency
- the remainder:
  - tool execution (~40% of active time)
  - setup env and initialization (29–45%)

What Tool Dominates?

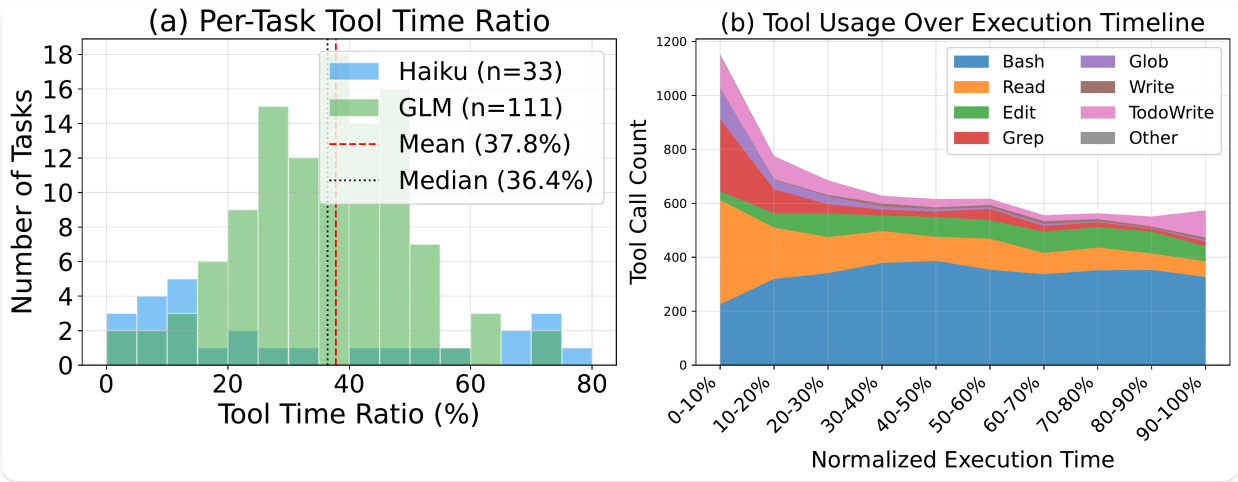
Q: Which tool types consume the most execution time?



- Bash dominates tool execution
- Top bash categories: test execution, Python snippet, package install

Tool Execution: Temporal Pattern

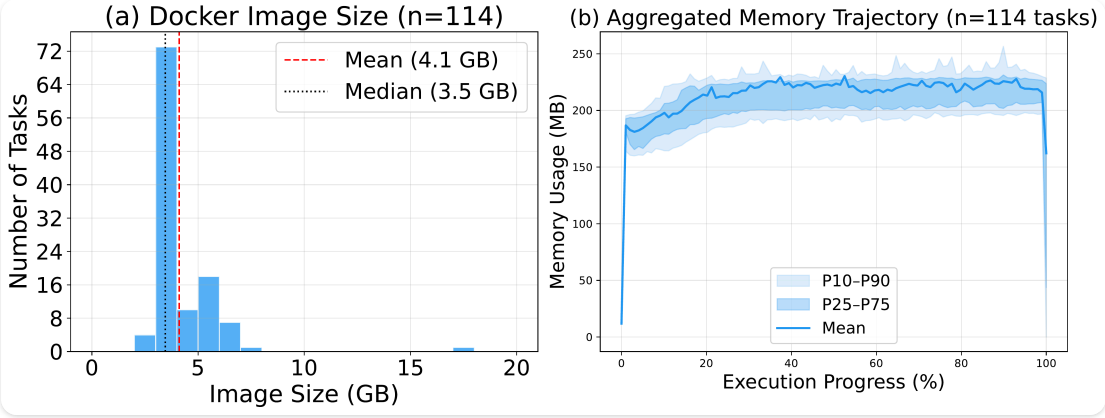
Q: When do different tools execute during a task?



- Tool calls follow an "**understand, modify, verify**" pattern: Read in first 30%, Bash peaks at 40–80%

Memory & Resource Structure

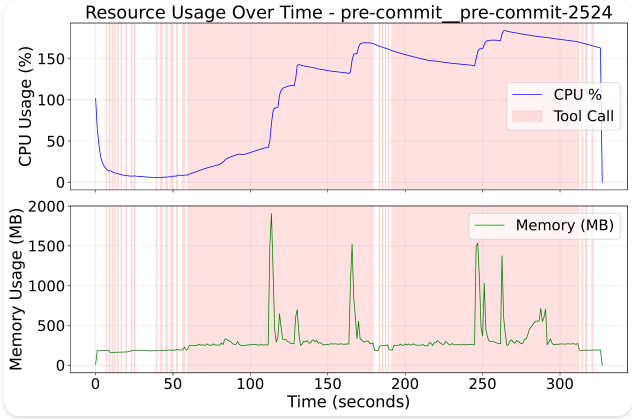
Q: What does resource consumption look like?



- **Memory, not CPU**, is the primary bottleneck (CPU avg <13%, memory peaks 2–4 GB)
- Resource consumption: **a ~185 MB framework baseline plus tool-call bursts**

Resource Time Series

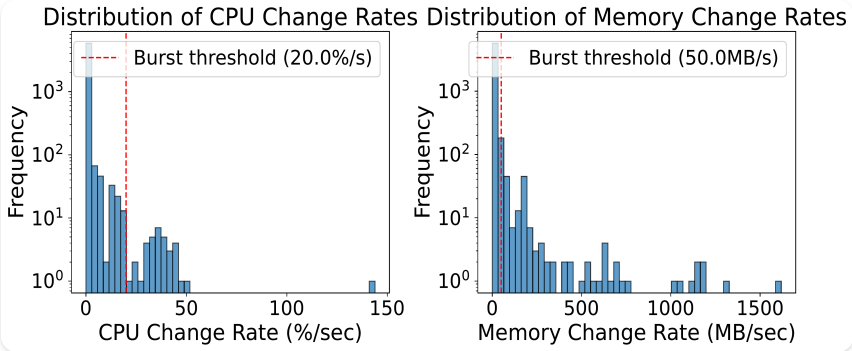
Q: What drives resource bursts?

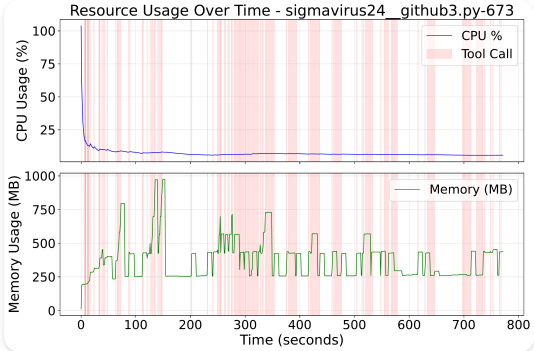


- Resource consumption is determined by what the tool *does* (e.g., pytest vs. git status), not which tool is invoked
  - **Bash calls differ by 13.7× in peak memory**
- **98.5% of memory bursts** occurring during **tool calls**

Burst Dynamics & Variability

Q: How fast and how variable are resource changes?





- Resource bursts last **1–2 seconds**
- peak-to-average ratio up to **15.4×**
- CPU-memory correlation varies by task **(−0.84 to +0.50); co-directional change cannot be assumed**

Characterization Summary

1. **OS execution dominates**: 56–74% of latency is tool execution + initialization, not LLM
2. **Memory is the bottleneck**: CPU avg <13%, but memory peaks 2–4 GB
3. **Demands are unpredictable**: 20× across tasks, 1.8× across runs; CPU/memory are decoupled

Implication:

- **System optimization** is important as LLMs gets faster!
- resource management for AI agents must operate at **tool-call granularity**

Agent vs. Cloud Workloads

Q: How different from existing cloud workloads?

Dimension	Serverless/FaaS	Microservices	Batch/HPC	**AI Coding Agent**
Execution duration	100ms–2s	Long-running	Minutes–hours	**5–11 minutes**
Container image	~50 MB	100 MB–1 GB	1–10 GB	**2.9–17.3 GB (med. 3.5)**
Memory footprint and peak/avg	128–512 MB, ~1.5×	Steady ~1 GB, 2–3×	Scales with data	**185 MB idle, peaks 2–4 GB, 15.4×**
CPU utilization	Brief spike	10–40%	80–100%	**<13% avg, peaks >1.7x**
Determinism	Deterministic	Mostly deterministic	Deterministic	**1.8× in same task**
Termination cost	Just retry	Can migrate	Lose progress	**Lose progress**

Three Mismatches

Q: Why do existing resource management tools fail?

	Static Limits	Reactive Control	Predictive Scaling
Tools	mem.max/high, cpu.max; K8s QoS	PSI; oomd; TMO	VPA; Autopilot
Assumes	Known peak; stable demand	Gradual pressure; kill OK	Repeatable; history valid
Agent	15.4× peak/avg; tool-semantic	1–2s burst; unpredictable	1.8× variance; kill = lose context
Mismatch	**Granularity**	**Responsiveness**	**Adaptability**

- **Granularity**: Not fine-grained, `memory.max` at peak wastes >90%; at average → OOM
- **Responsiveness**: Not fast enough, 3 GB/s bursts too fast for user-space tools (tens of ms reaction)
- **Adaptability**: Not semantic understanding, don't know what agent is doing

Agentcgroup Design and Prelimary Eval

- Fine-Grained Resource Domains to Seperate the framework from tool call(Bash wrap);
- Using eBPF for in-kernel enforce(Cgroup hooks);
- Intent-Driven Resource Allocation: agent declares what they need via config

Replay trace to evalution：

- Allow multi-tenant agents less OOM
- Programblely throttle → freeze
- Overhead negligible

Limitations & Future Work

Limitations

- Requires patched kernel for prototype (`memcg_bpf_ops` under review)
- Limited Evaluated
- Agents and tools on one machine

Future Work

- A LLM inside OS: let it understand what the agent does and negotate with workload
- Seperate the tool env and agent framework into different machine (like OpenClaw)
- Eval on more workoad

Thank You

Questions?

References

AgentCgroup Paper (arXiv:2602.09345) · AgentCgroup GitHub